IMPERIAL COLLEGE LONDON

DEPARTMENT OF ELECTRICAL AND ELECTRONIC ENGINEERING EXAMINATIONS 2023

OPTIMISATION

Thursday 4 May 10:00 am

Time allowed: 3:00 hours

There are FOUR questions on this paper.

Answer FOUR questions.

All questions carry equal marks

Any special instructions for invigilators and information for candidates are on page 1.

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OPTIMISATION

1. a) Consider the problem of minimizing a function $f : \mathbb{R}^n \to \mathbb{R}$. Assume that the function is C^1. Let x_k be the point obtained at the k-th iteration and determine the point x_{k+1} by minimizing the quadratic approximation of f given by

$$\tilde{f}(x) = f(x_k) + \nabla^\top f(x_k)(x - x_k) + \frac{1}{2t}(x - x_k)^\top (x - x_k),$$

with $t > 0$. Show that the resulting iteration coincides with the gradient method. [4 marks]

b) The *Envelop Theorem* for unconstrained optimization is formulated as follows.

> Let $f(x;a)$ be a function of the n variables $x_1, \ldots, x_n$ and of the parameter $a \in \mathbb{R}$. For each value of a, let $x^\star(a)$ be a stationary point of $f(x;a)$. Then
>
> $$\frac{df(x^\star(a);a)}{da} = \left.\frac{\partial f}{\partial a}\right|_{x=x^\star(a)}.$$

Consider the function

$$f(x;a) = -x^2 + 2ax + 4a^2.$$

Show that the envelope theorem holds. [4 marks]

c) Consider the optimization problem

$$\min_{x_1,\, x_2} \; -2x_1 - 3x_2,$$
$$\sqrt{x_1} + \sqrt{x_2} = 1,$$

with $x_1 \in [0,1]$ and $x_2 \in [0,1]$.

i) By plotting the admissible set and the level lines of the cost function determine the solution of the problem. [4 marks]

ii) Write necessary conditions of optimality for the problem and determine candidate optimal points (disregard the constraints $x_1 \in [0,1]$ and $x_2 \in [0,1]$ in the construction of the Lagrangian function).

Show that the optimal point determined in part i) does not satisfy the necessary conditions of optimality and explain why this is the case. [4 marks]

iii) By solving the equality constraint, transform the problem into an unconstrained problem and determine all local and global minimizers and maximizers for the problem. Compare the obtained result with the results obtained in parts i) and ii). [4 marks]

2. a) Consider the problem of maximizing the production of a company, with production function described by the Cobb-Douglas model

$$f(K,L) = AK^{\alpha}L^{\beta},$$

where K is the quantity of capital invested, L is the labour input, and α, β and A are positive constants. Let $p_K > 0$ and $p_L > 0$ be the cost of the unit of capital and of the unit of labour, respectively, and let $B > 0$ be the total production budget.

Assume for simplicity $\alpha = 1$, $\beta = 1$, and $A = 1$.

The problem can be therefore formulated as the problem of minimizing $-f$ subject to

$$p_K K + p_L L = B.$$

i) Write necessary conditions of optimality for the considered problem and determine candidate optimal solutions. [4 marks]

ii) Suppose now that the budget is increased by 10%, that is the new budget is $\tilde{B} = B + \delta B = \left(1 + \frac{1}{10}\right) B$, whence $\delta B = \frac{1}{10} B$. An estimate of the corresponding increase δf in the production function is given by

$$\delta f = \lambda^{\star} \delta B = \frac{1}{10} \lambda^{\star} B,$$

where $\lambda^{\star}$ is the optimal multiplier determined in part i). Compare this estimate with the exact value that one would obtain solving the problem with the constraint

$$p_K K + p_L L = 1.1B.$$

Comment on the accuracy of the estimate. [4 marks]

iii) The considered problem can be transformed taking the *log* of the cost to give

$$\min_{K,L} \; -\log(K) - \log(L)$$

$$p_K K + p_L L = B.$$

Using constraints elimination solve the transformed problem and show that the resulting solution coincides with the solution determined in part i). Show that this solution is indeed a global minimizer of the transformed problem, and hence of the original problem. [6 marks]

b) Consider now the more general production optimization problem described by the equations

$$\min_{q_1, q_2, \cdots, q_n} \; -U(q_1, q_2, \cdots, q_n)$$

$$p_1 q_1 + p_2 q_2 + \cdots + p_n q_n = B,$$

where $q_i \geq 0$ describes the quantity of goods/service of type i; $p_i > 0$ describes the average cost of goods/service of type i; U is a utility function (to be maximized, hence the $-$ sign in the formulation of the problem); and $B > 0$ is the available budget (to be used completely).

Show that the optimal solution and the corresponding multiplier are such that

$$\lambda^{\star} = \frac{1}{p_i} \frac{\partial U}{\partial q_i}(q^{\star}).$$

[6 marks]

3. Consider the optimization problem

$$P: \begin{cases} \min\limits_{x_1,x_2} x_1 x_2, \\ 2x_1^2 + x_2^2 \leq 1, \\ \\ x_1^2 + 3x_2^2 \leq 1. \end{cases}$$

a) By plotting the admissible set and the level lines of the cost determine the optimal solution of the problem. Show that all points are regular points for the constraints. [4 marks]

b) Write the first order necessary conditions of optimality for the problem. [4 marks]

c) Using the conditions in part b) determine candidate optimal solutions. [4 marks]

d) Classify the candidate optimal solutions computed in part c) and determine the global minimizer of the problem. Is this unique? [4 marks]

e) Consider the modified problem

$$P_m: \begin{cases} \min\limits_{x_1,x_2} x_1 x_2, \\ 2x_1^2 + x_2^2 = 1, \\ \\ x_1^2 + 3x_2^2 = 1. \end{cases}$$

Explain how one could use the results of parts b), c), and d) to determine candidate optimal solutions. Show that all admissible points for problem P_m are regular points for the constraints. [4 marks]

4. Consider the unconstrained optimization problem

$$\min_{x_1,\, x_2} \; 4x_1^2 - x_1x_2 + 2x_2^2 + 15x_1 + 30x_2.$$

a) Show that the point $P_\star = \left(-\frac{90}{31}, -\frac{255}{31}\right)$ is the global minimiser of the function by finding its stationary point and by showing that f is radially unbounded. Compute also the Hessian matrix Q of the function f. [2 marks]

b) Run one iteration of the gradient algorithm with exact line search with initial point $P_0 = (0,0)$ to determine a new point P_1. Let d_0 be the research direction used in this step. [4 marks]

c) Compute a direction d_1 which is Q-conjugate to the direction d_0 determined in part b). [4 marks]

d) Perform an exact line search along the direction d_1 determined in part c) starting from the point P_1 determined in part b) to determine the point P_2. Show that P_2 is the global minimizer of f and explain why this is the case. [4 marks]

e) An alternative way to solve the considered problem is to write an iteration based on the following considerations: $x_{1,k+1}$ is the minimiser of f with respect to x_1 for a fixed x_2, and $x_{2,k+1}$ is the minimiser of f with respect to x_2 for a fixed x_1.

Write explicitly the resulting iteration and run four steps of the iterations starting from P_0 in part b). [4 marks]

Compare the obtained results with the result obtained running the conjugate direction method and identify one advantage and one disadvantage of this method. [2 marks]